

SIBU, Malaysia

WMO: 964210

Lat: 2.262N

Lon: 111.985E

Elev: 122

StdP: 14.63

Time Zone: 8.00 (E08)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	71.6	72.4	69.2	107.9	80.3	69.8	110.1	78.9	14.0	81.3	12.2	82.0	3.3	140	0.287

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
5	15.3	93.8	79.0	93.2	79.0	91.9	78.8	81.5	89.0	80.9	88.5	80.3	87.9	4.9	270

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
79.2	152.4	84.8	79.0	151.2	84.5	78.6	149.2	84.0	45.3	89.5	44.6	89.3	44.1	88.6	91.6

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 11.7	9.8	8.4	DB	69.9	96.1	1.5	1.1	68.8	96.9	67.9	97.5	67.0	98.2	65.9	99.0	(4)
(5)			WB	68.6	84.1	1.5	2.5	67.6	85.9	66.7	87.4	65.9	88.8	64.9	90.6	(5)

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	81.7	80.4	80.8	81.8	82.4	82.8	82.3	82.3	82.2	81.8	81.5	81.5	81.1
	DBStd	2.00	1.91	1.85	1.79	1.79	1.94	2.05	2.07	2.14	1.97	1.81	1.48	1.65
	HDD50	0	0	0	0	0	0	0	0	0	0	0	0	0
	HDD65	0	0	0	0	0	0	0	0	0	0	0	0	0
	CDD50	11586	943	861	986	973	1017	970	1001	997	955	976	944	963
	CDD65	6111	478	441	521	523	552	520	536	532	505	511	494	498
	CDH74	52594	3613	3480	4433	4656	5196	4846	4968	4827	4387	4199	4001	3988
	CDH80	19575	1094	1123	1629	1770	2052	1920	2067	1992	1701	1524	1381	1322
Wind	WSAvg	3.5	3.5	3.7	3.6	3.4	3.3	3.3	3.4	3.5	3.5	3.4	3.4	3.4
Precipitation	PrecAvg	140.8	18.9	13.8	13.7	10.4	9.1	7.9	6.8	8.8	8.5	11.1	14.1	17.8
	PrecMax	194.8	34.9	26.6	25.5	17.0	18.4	14.6	15.7	17.5	19.6	20.9	27.3	28.3
	PrecMin	102.4	7.4	3.1	6.6	4.5	5.2	1.5	0.8	2.0	3.0	4.1	6.2	8.0
	PrecStd	23.0	7.3	5.4	4.7	3.2	3.2	3.7	3.7	3.9	3.7	3.9	5.3	4.7
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	91.5	91.8	93.4	93.6	94.8	94.8	95.0	95.1	93.6	93.4	91.9	91.7
		MCWB	78.7	78.4	78.7	79.9	79.7	79.2	78.8	78.1	79.3	78.5	78.9	79.0
	2%	DB	89.6	89.9	91.6	92.5	93.2	93.0	93.2	93.3	92.1	91.4	90.6	90.0
		MCWB	78.5	78.3	78.7	79.6	79.6	79.0	78.4	78.2	78.9	78.6	78.8	78.7
	5%	DB	87.9	88.3	89.9	91.1	91.6	91.4	91.6	91.6	91.0	89.8	89.4	88.7
		MCWB	78.6	78.2	78.5	79.4	79.6	78.9	78.2	78.2	78.7	78.4	78.7	78.6
	10%	DB	86.1	86.7	88.1	89.3	89.8	89.6	89.8	89.8	89.2	88.0	87.6	87.3
		MCWB	78.1	78.0	78.4	79.2	79.6	78.8	78.1	78.2	78.6	78.5	78.6	78.5
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	80.8	80.8	81.0	82.3	82.0	81.6	81.2	81.0	81.9	81.0	80.9	80.9
		MCDB	87.2	87.5	88.3	89.9	89.7	89.2	89.4	89.3	89.2	88.2	87.8	88.1
	2%	WB	79.8	79.8	80.1	81.0	81.1	80.7	80.1	80.1	80.3	80.0	80.0	79.9
		MCDB	86.2	86.5	87.4	88.5	88.9	88.7	88.4	88.5	88.2	87.3	87.0	86.8
	5%	WB	79.1	79.1	79.5	80.1	80.5	79.9	79.5	79.4	79.6	79.2	79.3	79.2
		MCDB	85.2	85.5	86.6	87.4	88.3	87.7	87.7	87.6	87.3	86.6	86.2	85.8
	10%	WB	78.5	78.4	78.9	79.5	79.8	79.2	78.8	78.8	78.9	78.7	78.8	78.7
		MCDB	84.2	84.5	85.6	86.4	87.2	86.6	86.9	86.8	86.5	86.0	85.6	85.0
Mean Daily Temperature Range	5% DB	MDBR	12.8	13.4	14.9	15.2	15.3	15.2	15.9	15.9	15.2	14.6	14.4	13.7
		MCDBR	15.2	15.6	16.9	17.3	17.5	17.6	18.1	18.5	17.7	17.0	16.2	15.7
		MCWBR	5.9	5.7	6.0	6.3	6.3	6.2	6.2	6.4	6.7	6.4	6.2	5.9
	5% WB	MCDBR	13.7	14.0	15.4	15.8	15.9	16.0	16.7	16.7	16.3	15.3	14.9	14.3
		MCWBR	5.9	5.7	6.1	6.5	6.4	6.5	6.7	6.7	7.0	6.5	6.3	6.1
Clear-Sky Solar Irradiance	taub	0.412	0.424	0.429	0.411	0.413	0.413	0.423	0.502	0.518	0.511	0.418	0.404	
	taud	2.466	2.409	2.392	2.440	2.444	2.443	2.407	2.170	2.129	2.149	2.439	2.488	
	Ebn at Noon	290	288	285	284	276	272	271	255	257	260	286	291	
	Edh at Noon	36	39	40	37	36	35	37	48	51	50	37	35	
All-Sky Solar Radiation	RadAvg	1367	1471	1598	1623	1562	1533	1568	1573	1483	1448	1453	1415	
	RadStd	115	83	73	67	69	80	75	86	119	133	60	71	

Historical Trends

	DBAvg	Heating			Cooling			Degree-Days			
		99% DB		99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
(46) Station Only	N/A	N/A	N/A	N/A	+0.50	+0.61	+0.45	N/A	N/A	+146	+146
(47) Regional (2 neighbors)	N/A	N/A	N/A	N/A	+0.72	+0.72	+0.49	N/A	N/A	+169	+160

Nomenclature: See separate page